

LotLine

A development feasibility navigator that catches conflicting public records before anyone acts on a tax-sale lot

Challenge 1 (Development Feasibility & Pro Forma Navigator) Startup track Build: Sat Sep 26, 9:00am to Sun Sep 27, 11:59pm ET
Plan v7 (final GO), Sept 24, 2026

Pitch. The Pittsburgh Land Bank’s Task Force says fragmented property data “continue[s] to limit strategic planning.” We checked the real October 2, 2026 City Treasurer Sale and found the problem in plain sight. The open data feed lists 96 parcels, but the City advertises 77. One “vacant” lot is 1,672 sq ft in the assessment record and 4,305 sq ft in County GIS, straddling its district minimum. Three “vacant” lots still have active condemned/dead-end cases attached, so their current condition is unverified. LotLine turns each parcel into a cited screening packet with a small, transparent Development Ease score and an evidence-coverage measure. It **detects source conflicts and refuses to score through them**, so no LLM prose can smooth them over.

1 Review history and verdict

Two judge agents over two rounds, then three Codex reviews. Round 3 verdict: **GO**. Rounds 4 and 5 confirmed every fix as resolved and ended with a final **GO** at about 65% top-3 odds; the remaining risk is implementation breadth, not the plan or data. The four fixes:

1. The “buildable envelope” is renamed to an *illustrative base-setback screen* showing both the interior-lot and corner-lot cases. Polygon checks flagged Michigan 15S66, Benezet and Dearborn as possible corner lots.
2. The conflict policy is now based on decision impact (critical, material, disclose) instead of a 50% cutoff.
3. The condemned conflict is worded as “current site condition unverified”.
4. The advertisement is reconciled by normalized PIN, with the upset price as a cross-check (77/77 on both).

New findings from the polygon checks:

- zoning has no split districts;
- slope over 25% and undermining overlaps on several lots, which moves Michigan 15S66 down to Conditional;
- a historic district on the Garfield Ave lot;
- FEMA zone A on two lots;
- a Registered Community Organization (RCO) for most lots.

The positive hero is now **Benezet St**. The full spec is frozen in `build_contract.md`.

2 Problem (every claim phrased as the source supports it)

- 13,770 parcels need intervention. A PLB Task Force *modeled* what it would take to process about 2,790 a year: more than \$32M annually. It identified unreliable property data and abandonment markers as a constraint (preliminary figures, Jan 2026).
- At the Treasurer Sale, buyers get only the taxing bodies’ interest. Mortgages, judgments, water claims and other secured claims survive, and there is a 90-day redemption period. The City’s rules say: “NO ONE SHOULD BID . . . WHO HAS NOT HAD THE TITLE EXAMINED.” Screening errors are expensive.
- Zoning relief is a recurring development friction. In 2026 the ZBA denied two-unit use variances on vacant lots in both H and RID-H (Kendall St, Rockland Ave).

3 User

A **public-interest acquisition analyst** at a land bank, the URA, a CDC or a City agency who screens a tax-delinquent sale list before committing title work. The Treasurer list is a *public proxy* for one segment of the tax-delinquent pipeline. It is not the Land Bank’s priority-bid list, which is Sheriff’s Sale only.

Every packet shows an **acquisition-route badge**: *City Treasurer Sale, competitive, 90-day redemption, title not cleared, no PLB priority verified*.

4 Outputs

- **Seven output values:** five screening states (Advance to staff review; Potential side yard or stewardship; Defer: missing or conflicting records; Defer: site conditions unknown; Do not advance under stated screening policy) and two routing states (out of sale universe; structure). *Advance* is worded as “apparent lower-discretion zoning path”, never “buildable”.
- **Development Ease score (0 to 6)** is always shown with its components (use entitlement, dimensional fit, checked environmental layers). The environment component counts three hazard families (terrain = landslide-prone or slope25; undermining; FEMA SFHA). Dimensional fit is 2 at 20 ft or more of envelope width, 1 at 10 ft to under 20 ft, and 0 below 10 ft. The total becomes a range when corner status is unverified, and “Partial: x of 4 known” when a component is withheld. The band is shown only when all components are known.
- **Separate evidence coverage:** 5 Boolean groups (parcel identity and area; encoded zoning rules; screening layers queried; enforcement records queried; acquisition route captured). Conflicts are reported as confidence, not as missing evidence.
- **Score rules:**
 - Unknown facts never score zero.
 - A *critical* conflict (current condition or sale universe) makes the parcel not scorable.
 - A *material* conflict withholds only the affected component.
 - A gap above 10% is disclosed without changing the score.
 - Title, legal access, utilities, survey and community alignment stay unscored flags.
 - No ranking across neighborhoods.
 - “No issue in checked layers” never means “clear”.
- **Source conflicts detected** (the headline feature):
 - sale universe: open data vs City advertisement;
 - parcel area: assessment vs County GIS, disclosed over 10%, material when the gap crosses a controlling minimum;
 - current condition: assessment “vacant” vs an active condemned/dead-end case;
 - zoning: point vs polygon split;
 - staleness by source date.
- **Structures:** a routing row only. “Structure indicated; vacant-land model not applicable.” Condemnation and demolition records are shown for routing. The 63 advertised structures are the case for v2.

5 The demo path that must never fail

1. Reconciliation: 96 WPRDC candidates, 77 advertised on 9/16 (matched by PIN, upset price checked), and 19 unreconciled records, with the note “sale status can change by payment or court order.”
2. Route the 63 advertised structures out of the vacant-land model; 14 vacant lots remain. PIN search box.
3. **Benezet St, New Homestead (0131N00031):**
 - R1D-L, 5,500 vs 5,327 sq ft (sources agree), about 49 x 111 ft;
 - no overlap in the checked screening layers;
 - score “5 to 6 of 6”, because corner status is unverified;
 - four flag tiles, evidence coverage and next checks.
4. **Centre Ave, Terrace Village (0010S00005):**
 - a critical conflict: assessment “vacant” vs an active condemned/dead-end case (current condition unverified);
 - a material conflict: 1,672 vs 4,305 sq ft straddles the 2,400 sq ft minimum;
 - result: **Not scorable, deed, survey and site review required.**
5. Compare Benezet with Michigan 15S66 (Conditional, 3 to 4 of 6 because of slope25, undermining and a possible corner). The flags explain the difference.
6. The memo keeps the conflict unresolved. The claim checker blocks a generated sentence that picks one source, and the injected text has no effect.

6 Golden set (advertised vacant lots; full table in the prep pack)

Parcel	Zone	Assess vs GIS sf	Preliminary outcome	Note
Benezet St (131N31)	R1D-L	5,500 / 5,327	Advance; 5 to 6 of 6	Clean-record example (not an acquisition pick); possible corner
Michigan St (15S66)	R2-H	3,000 / 3,141	Advance; 3 to 4 of 6 Conditional	slope25 + undermined; possible corner
Michigan St (14N100)	R2-H	2,562 / 2,633	Advance; 4/6 Conditional	27 ft; undermined; demolition lien
Dearborn St (50K227)	R1D-H	2,000 / 2,019	Advance, single-unit; 4 to 5 of 6 (band spans two)	20 ft; possible corner; Rockland card
Centre Ave (10S5)	RM-M	1,672 / 4,305	Not scorable	Critical + material conflict
Centre Ave (10R108)	LNC	2,675 / 2,129	Not scorable	Current-condition conflict
Walcott St (42D39)	RIV- RM	3,000 / 2,991	Not scorable	Current-condition conflict; two-unit only
Kemper St (88G313A)	P	10,276 / 4,606	Defer: rules not encoded	55% gap disclosed; P dimensions not encoded
Wylie Ave (10L127)	LNC	3,115 / 3,590	Defer: rules not encoded	Ch. 904; undermined; 7 casefiles (56 rows)
Mossfield, Platt, Banksville	H	agree on minimum	Defer: site	Slope standard 911.04.A.69; Kendall card
Saline St (88R1)	P	21,780 / 21,095	Defer: rules not encoded	Single-unit permitted in P; terrain flag
McClure Ave (75S108)	UI	4,680 / 4,593	Do not advance for housing	No housing use permitted

7 AI role and evaluation

The deterministic engine owns the use paths, dimensional fit, score, outcome and conflict detection. The LLM:

- writes the memo as {claim, fact_ids[]};
- explains conflicts and which decisions they affect;
- extracts “relief requested” for the two precedent cards, checked against hand labels.

The claim checker validates:

- numbers and categories against facts;
- code sections against a versioned allowlist;
- status words against the engine;
- that no conflict is resolved in prose;
- that pending law uses conditional wording;
- that approximations carry warnings;
- that no citation points at the wrong parcel.

Metrics:

- supported atomic claims;
- citation precision and coverage;
- engine fidelity;
- invalid section rate;
- correct abstention on the 10 adversarial and regression cases in the contract (real conflicts plus synthetic injection and stale-source fixtures);
- injection success rate;
- PII leakage;
- invariance to owner or neighborhood labels.

Neighborhood deferral statistics are dropped because the sample is too small. Instead we report deferrals by reason, missingness by source, and outcome changes caused by conflicts.

8 Scope

Keep:

- reconciliation;
- the four outcomes;
- single-unit and two-unit logic for R1D, R1A, R2, RM, H, P and UI, with abstention for LNC and RIV dimensions;
- parcel-polygon zoning and hazard intersections (FEMA, landslide);
- conflict detection;
- unique PLI casefiles;
- two ZBA cards;
- the score plus evidence coverage;
- the memo and claim checker;
- the injection and conflict tests;
- cached data only.

Cut:

- the consolidation remedy (no verified adjacent parcel);
- LLM remedy ranking;
- neighborhood charts;
- comps;
- full packets for nonresidential districts;
- more than two ZBA cards;
- the Bill 2025-1545 lens (Held in Council) unless everything else is done.

9 Schedule (solo with Cursor)

When	Work
Thu (done)	Sale list pulled and reconciled (96/77/14). Treasurer rules extracted. GIS areas and widths for all vacant lots. PLI casefiles aggregated. Condemned cross-check. Rule matrix corrected. ZBA cards. Source register.
Fri (no code, about 2h)	Mostly done in <code>build_contract.md</code> . Remaining: check corner and frontage for Benezet, Michigan and Dearborn on the County portal or aerial imagery; hand-verify the §911 cells for P, LNC and RIV-RM; read the contract once and take the fallback screenshots on Sunday.
Sat 9–13	Vertical slice first: load the prepared CSVs, then build Benezet and Centre end to end (facts, tiles, score, conflict, memo stub).
Sat 13–20	Generalize to the 14 vacant lots and the reconciliation table, add the claim checker and the acceptance test from <code>build_contract.md</code> . Refresh the snapshot only if time allows.
Sat 20–24	Memo, claim checker, precedent cards, export. Full rehearsal before sleeping.
Sun 9–12	Eval and adversarial runs. Freeze parcels and metrics at noon.
Sun 12–15	README (sources, AI disclosure, real vs approximate vs mocked). Code freeze at 3pm.
Sun 15–19	Re-check the City advertisement and Legistar. Record and upload the video. Submit.

10 Demo script (4 minutes; fallbacks in `build_contract.md`)

- 0:00–0:15** The decision question and the pitch.
- 0:15–0:40** Reconciliation: 96 vs 77 vs 19.
- 0:40–1:00** 63 structures routed out; 14 vacant lots with outcomes.
- 1:00–1:45** Benezet: four tiles, 5 to 6 of 6, coverage 5/5, next checks. Framed as “market demand and appraisal not evaluated”.
- 1:45–2:35** Centre Ave refusal.

2:35–2:55 Benezet vs Michigan 15S66.

2:55–3:35 Memo, the blocked claim, the injection fixture.

3:35–4:00 Limitations; what is real, approximate or mocked; pilot and v2.

Dearborn and the Rockland card stay in the app and README only.

Sources

[WPRDC Treasury Sales](#); [City advertisement 9/16/2026](#); [Treasurer Sale regulations](#); [County parcels \(PASDA\)](#); [Condemned properties](#); [PLI violations](#); [PLB Task Force report](#); [PLB acquisition guide](#); [§903](#); [§905](#); [§911](#); [§921](#); [ZBA Kendall St](#); [ZBA Rockland Ave](#); [Bill 2025-1545](#).